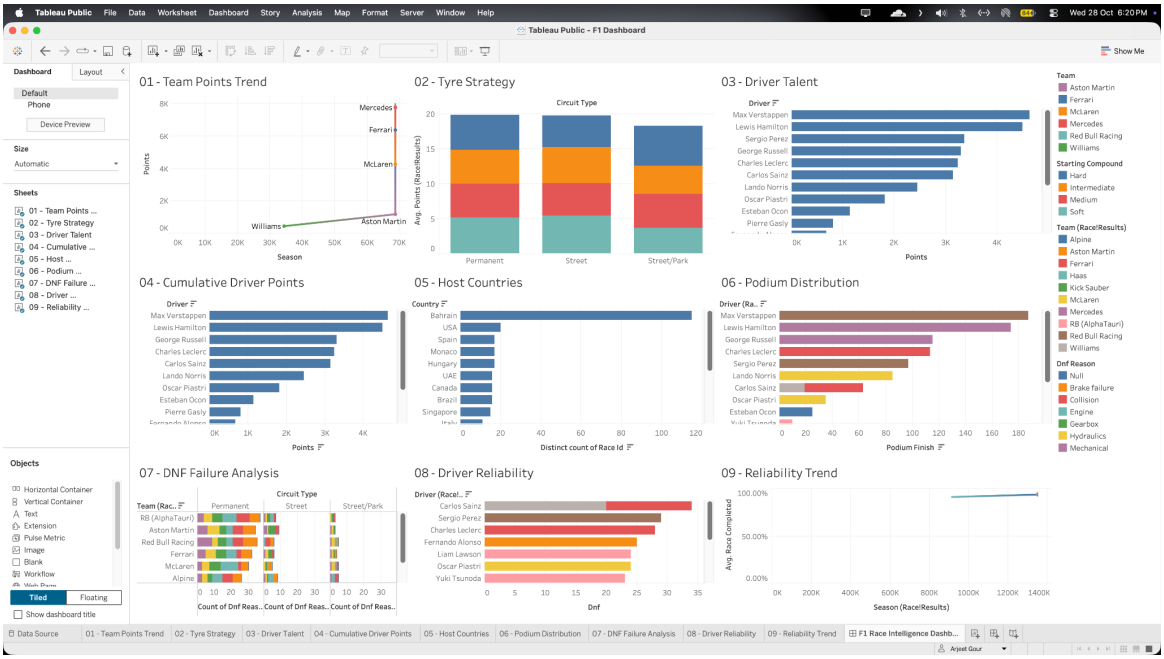
PROJECT REPORT

FORMULA 1
RACE INTELLIGENCE ANALYSIS

Interactive data analysis and visualisation using Tableau Public



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Project Guide	[To be added]

Submitted in partial fulfilment of the academic project requirements.

DOCUMENT STRUCTURE

Section	Title
1	Introduction
2	Ideation Phase
3	Requirement Analysis
4	Project Design
5	Project Planning & Scheduling
6	Functional and Performance Testing
7	Results
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Project summary. This project uses Tableau Public to turn an Excel-based Formula 1 dataset into a dashboard containing nine views across team performance, tyre strategy, driver results, host countries, podiums, DNFs and race completion reliability.

Data interpretation note. The workbook includes modern driver names in early seasons (for example, the standings sheet lists Max Verstappen in 2010). Therefore, the supplied workbook is treated as the project's analysis dataset, not independently verified historical F1 records. Findings should be described as patterns in the supplied dataset.

1. INTRODUCTION

1.1 Project Overview

Formula 1 is a data-intensive motorsport in which team results, driver performance, tyre choices, circuit characteristics and reliability all affect race outcomes. This project, *F1 Race Intelligence Analysis*, presents an interactive Tableau Public dashboard designed to explore these factors through visual analytics. Instead of requiring users to inspect large spreadsheets manually, the dashboard organises important indicators into nine charts that can be explored using filters, marks, and tooltips.

The supplied Excel workbook contains five sheets: **race_results**, **championship_standings**, **lap_intelligence**, **qualifying_results**, and **circuits**. The main race-results sheet has 6,464 rows and 42 fields. Across the supplied tables, the data covers seasons 2010–2026. The workbook records 334 distinct race IDs, 19 distinct drivers, 10 teams and 21 host countries in race_results.

1.2 Purpose

The purpose is to build a clear, interactive and stakeholder-oriented visual analysis of F1 performance. The dashboard is intended to help a team strategist compare team and driver outcomes, help a broadcast analyst retrieve leaderboard and host-country views, and help a technical analyst examine retirement reasons and completion trends. It also demonstrates practical use of Tableau for connecting spreadsheets, creating calculated fields, building visualisations and assembling a dashboard.

Project objectives

• Compare team championship points across seasons.
• Explore starting tyre compounds against circuit types and race points.
• Rank drivers using points and podium measures.
• Count Grand Prix rounds by host country without double-counting driver rows.
• Examine DNF reasons by team and circuit type.
• Measure driver-race completion rate and view its trend by season.
• Publish a dashboard for interactive exploration.

2. IDEATION PHASE

2.1 Problem Statement

Formula 1 produces large amounts of race, qualifying, lap, tyre, team and reliability data. Reviewing these records separately makes it difficult for stakeholders to quickly compare teams, identify useful driver-performance patterns, understand tyre outcomes across circuit types, count host-country race rounds, and detect recurring retirement reasons. The project addresses this problem by bringing key measures into one Tableau dashboard with nine visualisations and interactive filtering.

The dashboard is organised around three stakeholder scenarios:

Scenario	Stakeholder need	Dashboard support
1. Team performance strategist	Compare team points, tyre strategy outcomes and driver talent.	Team points trend; tyre strategy; driver talent leaderboard.
2. Broadcast analyst	Quickly retrieve driver points, host-country race counts and podium distributions.	Cumulative driver points; host countries; podium distribution.
3. Technical regulation analyst	Investigate retirement reasons, driver DNFs and completion-rate trends.	DNF failure analysis; driver reliability; reliability trend.

2.2 Empathy Map Canvas

Says	Thinks	Does	Feels
“I need a quick comparison of team and driver performance.”	Which patterns are meaningful and comparable?	Filters by season, team, driver and circuit type; inspects tooltips.	Wants confidence that figures are clear, consistent and trustworthy.

This empathy map is a design interpretation of the three fictional stakeholder scenarios supplied in the problem statement; it is not based on interviews with real users.

2.3 Brainstorming

The proposed ideas were narrowed to nine visualisations: (1) team points trend, (2) tyre strategy analysis, (3) driver talent, (4) cumulative driver points, (5) host countries, (6) podium distribution, (7) DNF failure analysis, (8) driver reliability, and (9) race completion reliability trend. A single dashboard was selected because it allows users to move between strategic, media and technical questions without switching tools.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User action	Need / pain point	Dashboard response
Discover	Open the dashboard	Needs an understandable overview	Clear title and grouped charts
Explore	Select a season, team or driver	Large data is difficult to scan	Filters and interactive marks
Compare	Compare points, tyre outcomes or DNFs	Measures can be misleading if aggregated incorrectly	Explicit chart titles, axes and tooltips
Interpret	Identify trends or outliers	Needs context and limitations	Season context and reliability measures
Share	Present results to others	Needs portable and accessible output	Tableau Public dashboard URL

3.2 Solution Requirement

Functional requirements: connect to the Excel workbook; access the five sheets; create nine worksheets; create calculated fields for podium finish and race completion; support season/team/driver/circuit-type filtering where relevant; show tooltips; assemble a single dashboard; publish the dashboard to Tableau Public.

Non-functional requirements: readable chart labels; consistent colour use; correct aggregation; responsive dashboard layout; understandable titles; reasonable rendering speed; and transparent explanation of data gaps and assumptions.

3.3 Data Flow Diagram

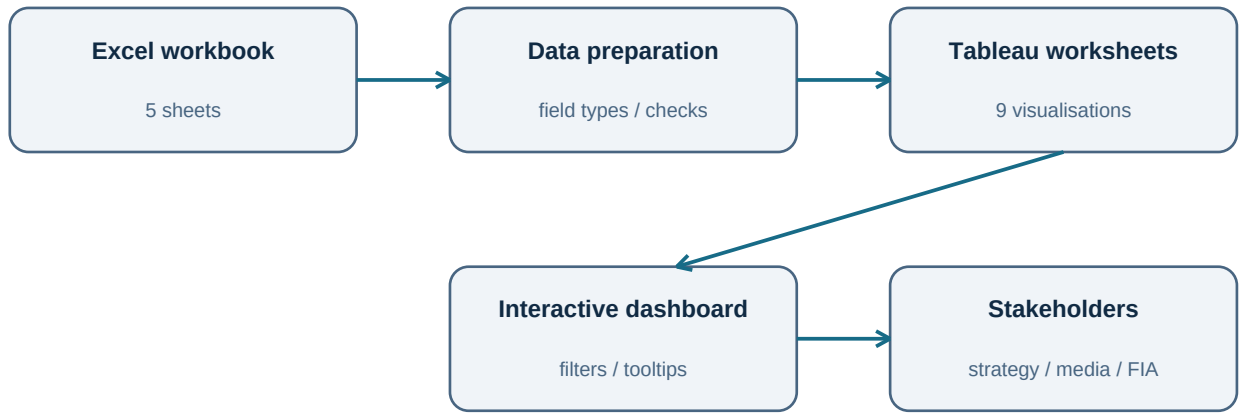


Figure 1. High-level data flow from the Excel workbook through Tableau preparation and worksheets to the dashboard and its stakeholders.

3.4 Technology Stack

Component	Use in the project
Microsoft Excel (.xlsx)	Source workbook with five sheets and structured race data.
Tableau Public Desktop	Connect to Excel, create calculated fields, charts, filters, tooltips and dashboard.
Tableau Public online	Publish and share the interactive dashboard.
PDF documentation	Record the project purpose, design, implementation, tests, results and limitations.

4. PROJECT DESIGN

4.1 Problem Solution Fit

The solution fits the problem because the stakeholders need comparative answers rather than raw rows of data. Line charts support season-by-season trends; ranked bars make leaderboards easy to scan; stacked bars reveal the composition of podiums or DNF reasons; and filters let users focus on selected seasons, teams, drivers or circuit types. Tableau Public provides a single shareable dashboard for these tasks.

4.2 Proposed Solution

No.	Visualisation	Purpose / measure
1	Team Points Trend	Line chart of team championship points by season; compare Red Bull Racing, Mercedes and other teams.
2	Tyre Strategy	Compare average race points by starting compound and circuit type; lap-level compound analysis can use lap_intelligence.
3	Driver Talent	Rank drivers by points, with wins and podiums in tooltips.
4	Cumulative Driver Points	Driver leaderboard over the selected seasons.
5	Host Countries	Distinct race IDs by country, using COUNTD(race_id).
6	Podium Distribution	Count driver finishes in positions 1–3, segmented by team or finish position.
7	DNF Failure Analysis	DNF counts by team, reason and circuit type.
8	Driver Reliability	Rank drivers by recorded DNF count; interpret alongside race starts.
9	Reliability Trend	Average Race Completed flag by season, formatted as a percentage.

4.3 Solution Architecture

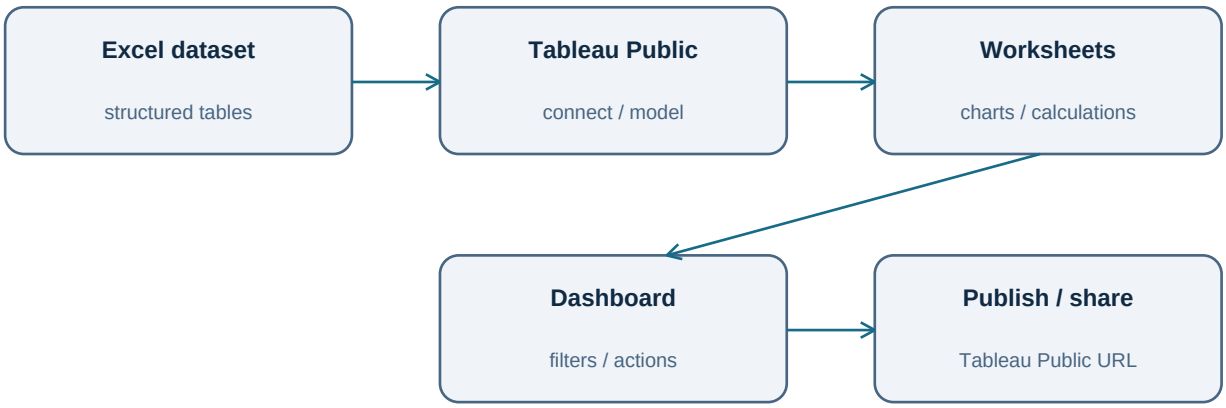


Figure 2. Proposed architecture: workbook → Tableau Public → worksheets → interactive dashboard → published link.

Data model and aggregation notes

The tables have different grains: race_results is driver-race-level, championship_standings is driver-season-level, lap_intelligence is lap-sample-level, qualifying_results is driver-qualifying-level, and circuits is circuit-level. They should not be blindly joined into one flat table because row multiplication can inflate totals. Use separate data sources or Tableau relationships based on matching keys and validate counts after linking.

Suggested calculated fields

Podium Finish:

```
IF [finish_position] <= 3 THEN 1 ELSE 0 END
```

Race Completed:

```
IF [dnf] = 0 THEN 1 ELSE 0 END
```

Race Completion Rate: use AVG([Race Completed]) at the intended grain, then format as a percentage. This avoids dividing an aggregated field by a row count unnecessarily.

For the team points trend, use season as a year/date dimension (not SUM(season)) and sum points at the intended team-season grain. For host countries, use COUNTD(race_id) rather than row count. For tyre analysis, average points is more comparable than total points when compound usage differs.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Planned task	Deliverable / completion evidence
1. Problem understanding	Identify three stakeholder scenarios and their questions.	Problem statement and objectives
2. Dataset review	Inspect sheets, field names, record counts and data gaps.	Dataset inventory and field understanding
3. Visual design	Select chart types and measures for each question.	Nine worksheet plan
4. Tableau implementation	Connect workbook, create calculated fields, charts and filters.	Nine worksheets
5. Dashboard assembly	Arrange charts, format, add interactions and test usability.	Interactive dashboard
6. Validation	Check counts, aggregations, axes, filters and edge cases.	Testing record and correction list
7. Documentation	Prepare report, add screenshot, link and limitations.	Final PDF report

A calendar with exact dates was not supplied; the phases above are therefore presented as a logical project schedule rather than claiming that tasks were completed on specific dates.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

The following test cases are recommended for validating the dashboard before final submission. The report distinguishes checks inferred from the current screenshot from checks that must be run in Tableau; it does not claim unperformed tests passed.

Test ID	Test / expected result	Status from supplied screenshot / workbook
T01	Workbook connection: all five sheets are available.	Workbook structure inspected; Tableau connection should be confirmed in the workbook.
T02	Nine worksheets are present on the dashboard.	Visible in supplied screenshot.
T03	Season axis displays individual years, not summed season values.	Correction needed: screenshot shows very large numeric season axes on charts 1 and 9.
T04	Host country chart counts distinct races, not driver rows.	Axis says distinct count of race ID; validate against COUNTD(race_id).
T05	Podium count includes only finish positions 1, 2 and 3.	Validate the filter and calculation in the Tableau worksheet.
T06	DNF view filters to actual retirements and groups by reason.	Chart is visible; check that Null reasons and non-DNF rows are excluded appropriately.
T07	Completion rate is between 0% and 100%.	Visible axis is 0%–100%; correct season dimension still needs verification.
T08	Season/team/driver filters update intended views without duplicating totals.	Must be tested interactively in Tableau.
T09	Dashboard opens from the published URL and is readable.	Link supplied by student; verify in a signed-out/incognito browser before submission.
T10	Dashboard responds smoothly to filters and selections.	No measured performance timing supplied; test on target device/network.

Suggested corrections before final submission

1. In charts 1 and 9, change season from SUM(season) to a discrete or continuous year dimension. The current screenshot indicates that season values are being summed, making the x-axis unsuitable for trend interpretation.
2. Confirm chart 1 aggregates points at team-season level, and that the data source does not duplicate standings rows.
3. Review chart 2's axis and title: the displayed measure is average points per race result, not a direct tyre-grip or stint-performance measure.
4. Ensure DNF analysis excludes non-DNF entries and decide how null DNF reasons should be handled.
5. Add clear notes that 2026 may be an incomplete season and that the workbook's early-season driver assignments do not align with real historical F1 chronology.

7. RESULTS

7.1 Output Screenshots

The screenshot below records the current Tableau Public dashboard implementation. It shows all nine worksheets arranged in a three-column layout with legends for team, starting compound, team, and DNF reason. It is included as evidence of the current visual design, not as independent proof that every calculation has been validated.



Figure 3. Screenshot supplied by the student: F1 Race Intelligence Dashboard in Tableau Public.

Dataset profile

Dataset item	Observed value
race_results	6,464 rows; 42 fields; 334 distinct race IDs.
championship_standings	311 rows; seasons 2010–2026; 19 drivers; 10 teams.
lap_intelligence	3,814 rows; includes lap time, sector time, tyre compound, tyre age, DRS and speed-trap fields.
qualifying_results	3,130 rows; includes qualifying positions, session lap times, tyre compound and top speed.
circuits	29 circuit records with country, city, laps, length and circuit type.
Race-result coverage	19 drivers, 10 teams, 21 countries, 17 seasons.
Recorded DNF flags	396 rows with dnf=1 (sum of supplied dnf flag).

Observations and limitations

The supplied race_results sheet contains 6,464 driver-race rows and 396 rows flagged as DNF. The recorded DNF reasons include Engine (73), Hydraulics (70), Mechanical (67), Brake failure (64), Gearbox (62) and Collision (60); these counts are values in the supplied workbook and should be checked against the dnf flag because the reason field may be populated or missing independently of the flag. Starting compounds include Soft, Medium, Hard and Intermediate. Race rows cover Permanent, Street and Street/Park circuit types.

The workbook does not include a driver nationality field. Therefore, the current dashboard cannot substantiate a ranking of driver nationalities or talent pools by nationality. To add that analysis, attach a verified driver-to-nationality lookup table. Also, a chart of raw DNF counts does not by itself establish which driver or team is most reliable; compare against race starts or

use completion rate.

The team points and driver leaderboard visuals can show the patterns present in the workbook, but this report avoids asserting a specific team or driver as the real-world season champion because the dataset has not been externally validated. The dashboard's Tableau Public URL is provided in the Appendix.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Combines nine analytical views in a single dashboard.
• Makes season, team, driver, tyre, country and reliability patterns easier to inspect.
• Supports interactive exploration through filters, selections and tooltips.
• Uses different chart types suited to different questions: lines for trends, bars for rankings, and stacked bars for category composition.
• Uses an Excel workbook, making the data source straightforward to update if the schema remains consistent.
• Published Tableau Public link makes the dashboard easy to share for academic review.

Disadvantages / limitations

• Tableau Public workbooks and published views are public; confidential data should not be used.
• Missing driver nationality means nationality analysis requires an additional verified dataset.
• Dataset chronology and driver/team labels require validation before making historical claims.
• Aggregations can be misleading if tables of different grains are joined incorrectly.
• Raw DNF counts are affected by the number of race starts; normalized rates are needed for fair reliability comparisons.
• A season-level dataset may include an incomplete latest season, making totals across years less comparable.
• The dashboard screenshot shows at least two season axes that need correction before final evaluation.

9. CONCLUSION

The F1 Race Intelligence Analysis project demonstrates how Tableau Public can transform a multi-sheet Excel workbook into an interactive dashboard for strategic, broadcast and technical analysis. The nine views address team championship trajectories, tyre strategy, driver performance, host-country race counts, podium distribution, DNF reasons, driver retirement counts and race completion trends. The screenshot confirms that the nine visualisations have been assembled into one dashboard. To make the final deliverable analytically reliable, the season axes should be corrected, calculations and filters validated, and the dataset's historical consistency clearly qualified. The project provides a useful foundation for exploratory motorsport analytics while recognising the limits of the supplied data.

10. FUTURE SCOPE

• Add a verified driver-nationality lookup and create nationality-level talent-pool views.
• Add an explicit race-start denominator and normalized DNF rate by driver, team and circuit type.
• Compare tyre compounds using lap-time and tyre-age data from lap_intelligence, controlling for circuit and weather where possible.
• Add pit-stop strategy, undercut/overcut attempts, safety-car periods, qualifying pace and positions gained to expand strategy analysis.
• Validate all race and championship records against an authoritative historical source before drawing real-world conclusions.
• Improve the dashboard with KPI cards, explanatory annotations, accessible colour choices and a dedicated page or story for each stakeholder scenario.
• Re-test performance and filter behaviour after each dataset refresh.

11. APPENDIX

A. Dataset and dashboard links

Dataset file: F1 Racing Project Dataset.xlsx (provided with the project).
Tableau Public dashboard: Open F1 Race Intelligence Dashboard
GitHub repository: [Not provided — add link if available].
Project demo/video: [Not provided — add link if available].

B. Source code

No separate source-code file was supplied. The core project is implemented in Tableau Public using worksheet configuration, calculated fields, filters and dashboard actions rather than a conventional standalone program. If the course requires source code, include any Tableau calculated fields and workbook file that the instructor requests.

C. Field and measure notes

Field / measure	Meaning / recommended handling
season	Year dimension. Do not aggregate by SUM; use as discrete year or continuous date/year.
points	Points at the row grain. Aggregate only after checking whether the data source has duplicate records.
race_id	Race identifier. Use COUNTD(race_id) to count unique Grand Prix rounds across driver-level rows.
dnf	Retirement flag. Use SUM(dnf) for count only if values are consistently 0/1 and each row represents one driver-race entry.
dnf_reason	Recorded retirement reason; check missing values and whether reason entries correspond to dnf=1.
starting_compound	Starting tyre compound in race_results. It is not necessarily the tyre used throughout the race.
Race Completed	Calculated as 1 when dnf=0, otherwise 0; AVG at driver-race grain yields completion rate.
Podium Finish	Calculated as 1 for finish positions 1–3, otherwise 0; sum to count podium entries.

D. Submission checks

Before submission, replace the project guide placeholder if required; confirm the Tableau Public link is published and accessible; correct season axes; validate chart aggregations and filters; and confirm whether the instructor requires a .twbx packaged workbook alongside this PDF.

— End of Report —